lothian_cluster

Guanyu

2026-01-22

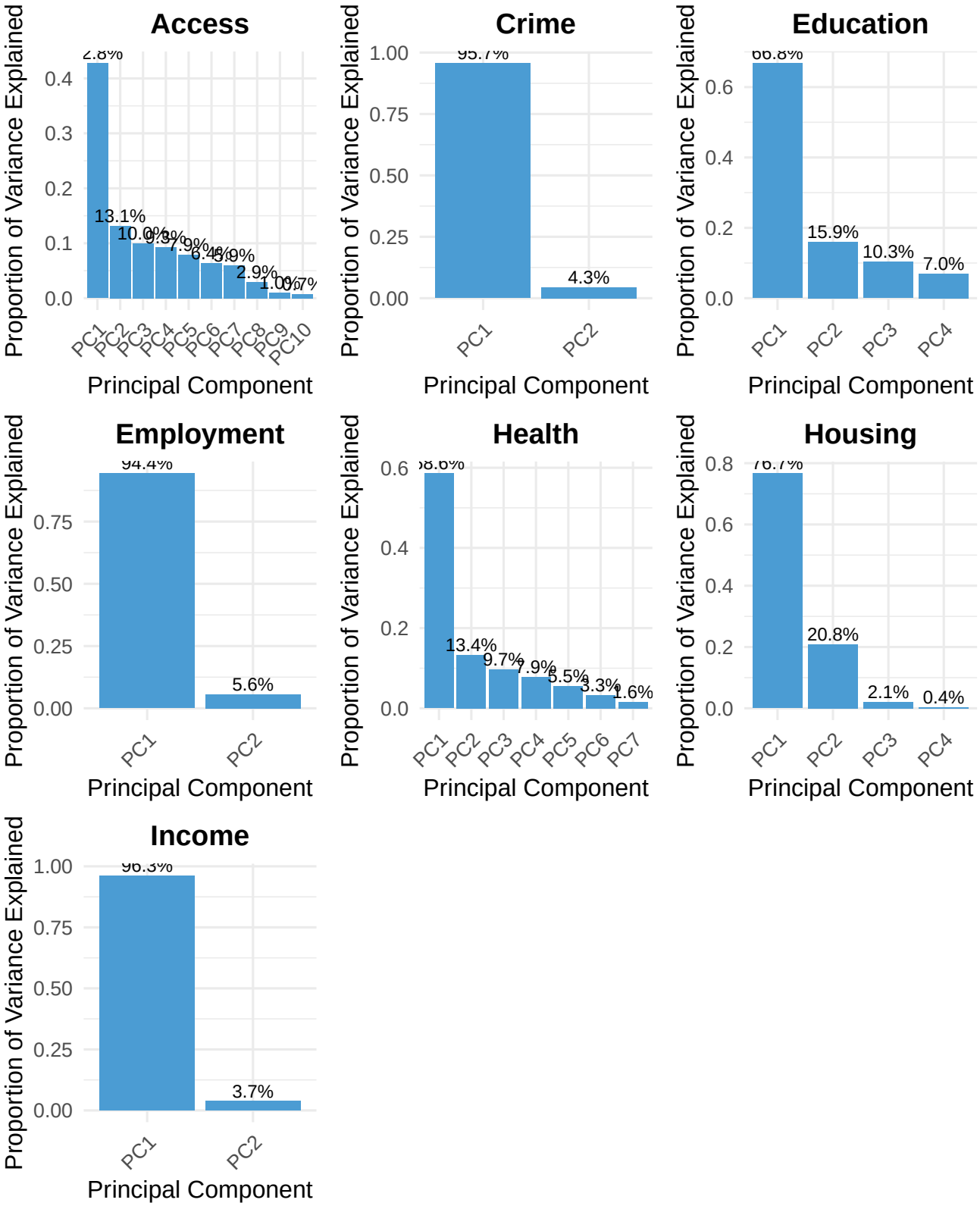
```
## Reading layer `sc_dz_11' from data source
##   `/Users/chenguanyu/Desktop/SIMD20/SIMD/simd2020_withgeog/sc_dz_11.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 6976 features and 9 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -8.650007 ymin: 54.63324 xmax: -0.7246094 ymax: 60.86077
## Geodetic CRS:  WGS 84
```

Table 1: Indicators with Missing Values in Lothian

Indicator	Missing_Count	Percent_Missing
Attendance	179	16.53
Attainment	82	7.57
crime_count	36	3.32
crime_rate	36	3.32

Domain−wise PCA Results for Lothian Indicators

Each plot shows variance explained by principal components within its domain



PCA Loadings for Health

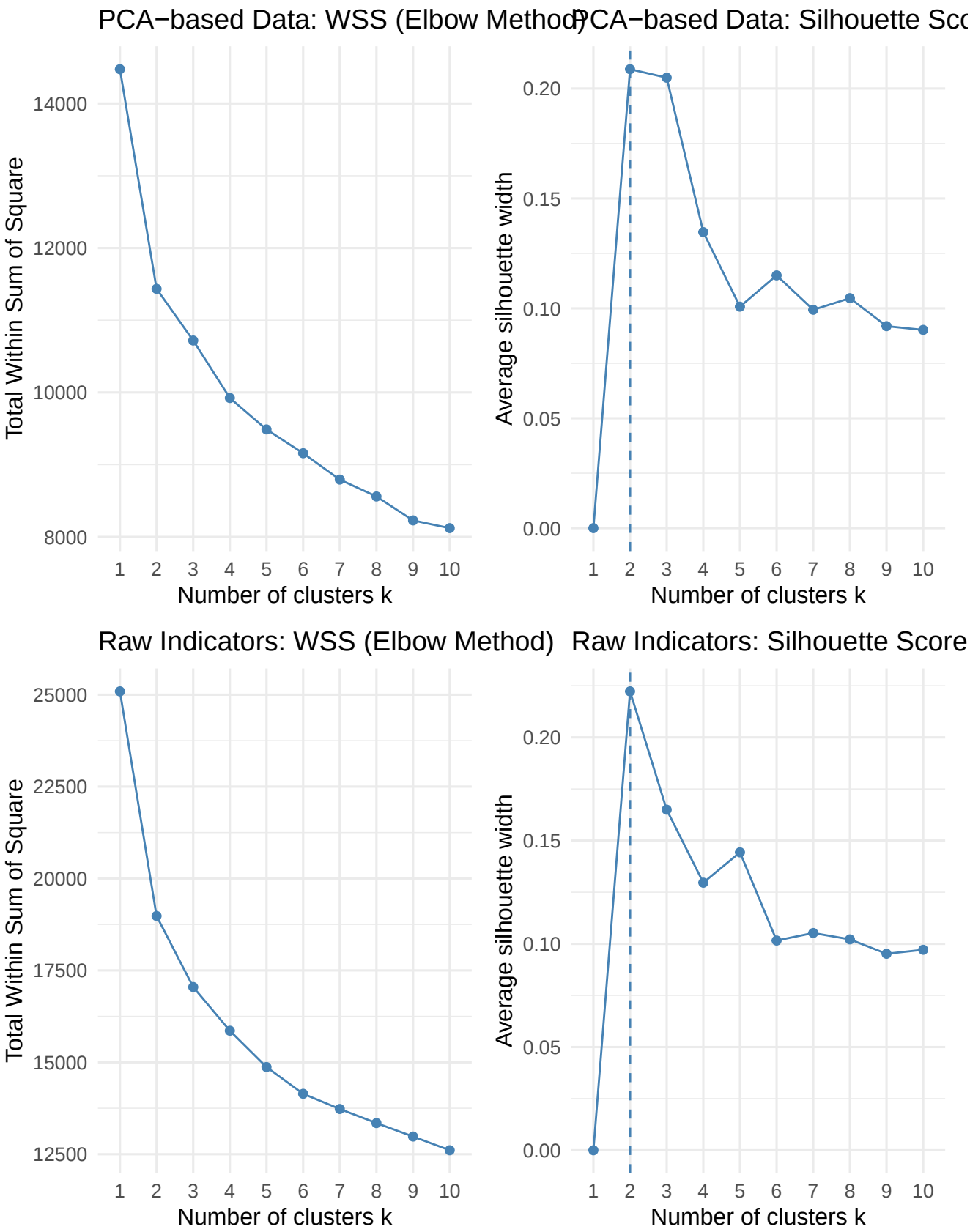
Variable	Health_PC1	Health_PC2	Health_PC3
CIF	-0.4652781	0.0266189	-0.0920688
ALCOHOL	-0.3843177	0.0070472	-0.3181707
DRUG	-0.3918426	0.0230941	-0.2374422
SMR	-0.3044407	0.2193520	0.8890277
DEPRESS	-0.4124313	0.0955564	0.0002658
LBWT	-0.1545163	-0.9692938	0.1864685
EMERG	-0.4422857	0.0439507	-0.0936545

PCA Loadings for Education

Variable	Education_PC1	Education_PC2
Attainment	0.5293641	-0.0506184
no_qualifications	-0.5425136	0.1481482
not_participating	-0.4368680	-0.8332377
University	0.4843542	-0.5302874

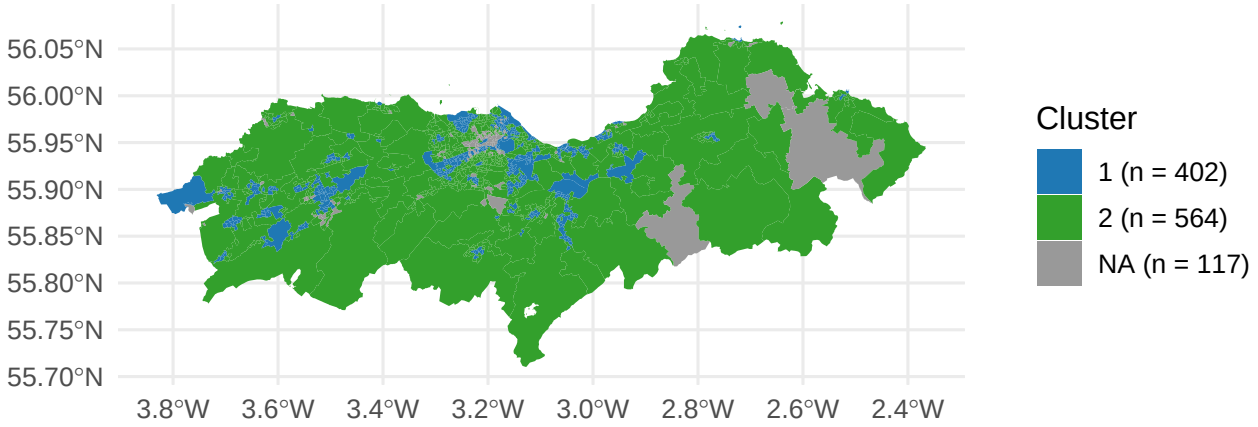
PCA Loadings for Access

Variable	Access_PC1	Access_PC2	Access_PC3	Access_PC4	Access_PC5
drive_petrol	-0.2228965	-0.2544601	-0.4946992	-0.4685466	0.1162793
drive_GP	-0.3653041	-0.0053158	0.1129203	0.3171167	0.4595718
drive_post	-0.3120565	0.5440056	-0.0696425	-0.3065276	-0.1330608
drive_primary	-0.2180980	0.1005098	-0.3033029	0.5253085	-0.6861377
drive_retail	-0.3306484	-0.4021670	0.3710941	-0.1456583	-0.2749781
drive_secondary	-0.2673019	-0.3272045	-0.3995557	-0.1591461	-0.0014754
PT_GP	-0.4026238	0.0898773	0.2042777	0.2425782	0.3288413
PT_post	-0.3472891	0.5107205	0.0634175	-0.2769277	-0.0444593
PT_retail	-0.3700482	-0.2959790	0.3839012	-0.0908241	-0.2328161
broadband	-0.2673837	-0.0621240	-0.3939444	0.3459292	0.2164513

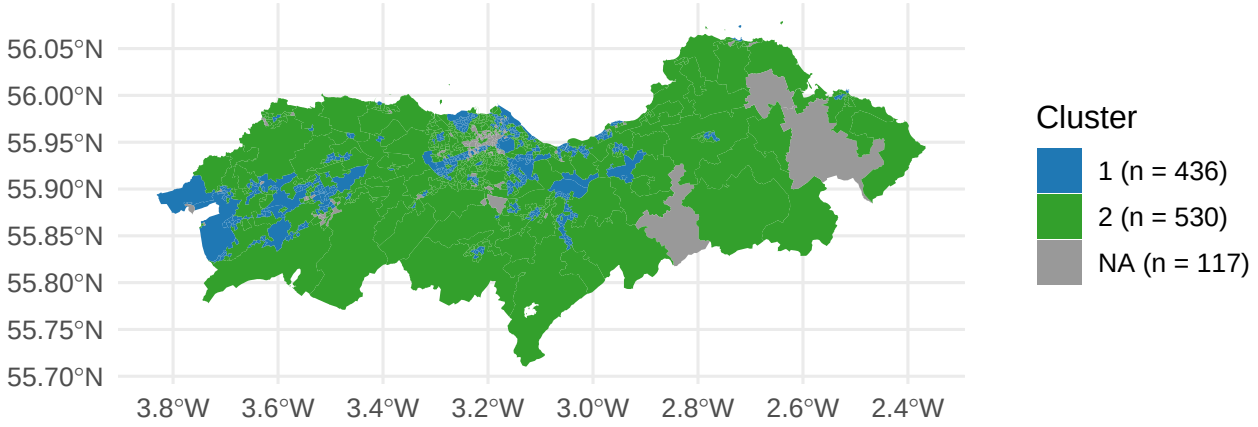

PCA−based Data: WSS (Elbow Method)
Total Within Sum of Square
14000
12000
10000
8000
1 2 3 4 5 6 7 8 9 10
Number of clusters k
PCA−based Data: Silhouette Sco
Average silhouette width
0.20
0.15
0.10
0.05
0.00
1 2 3 4 5 6 7 8 9 10
Number of clusters k
Raw Indicators: WSS (Elbow Method)
Total Within Sum of Square
25000
22500
20000
17500
15000
12500
1 2 3 4 5 6 7 8 9 10
Number of clusters k
Raw Indicators: Silhouette Score
Average silhouette width
0.20
0.15
0.10
0.05
0.00
1 2 3 4 5 6 7 8 9 10
Number of clusters k

Comparison of K−means Clusters (k = 2): PCA vs Raw

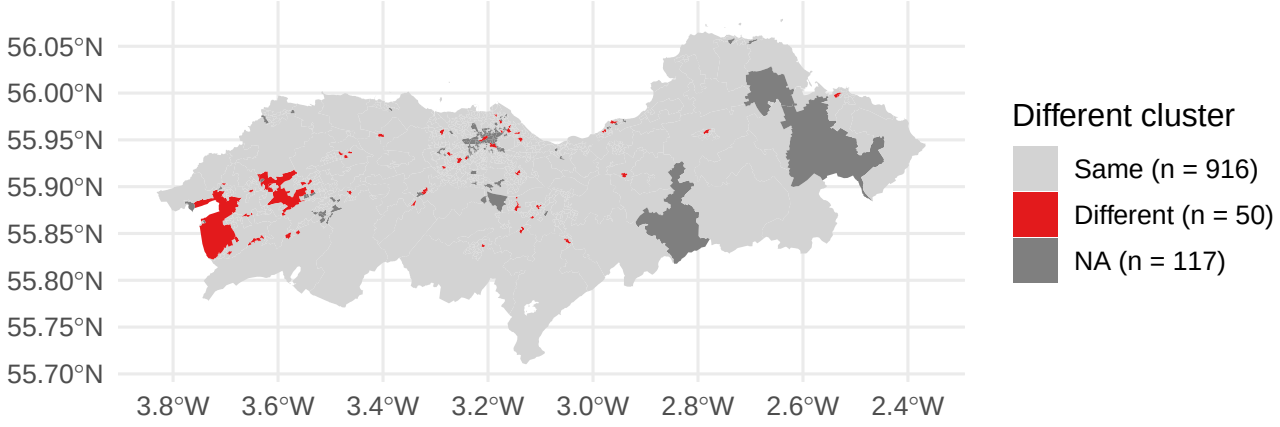
K−means Clusters (PCA−based, k = 2)



K−means Clusters (Raw indicators, k = 2)



Areas with Different K−means Assignments (PCA vs Raw)



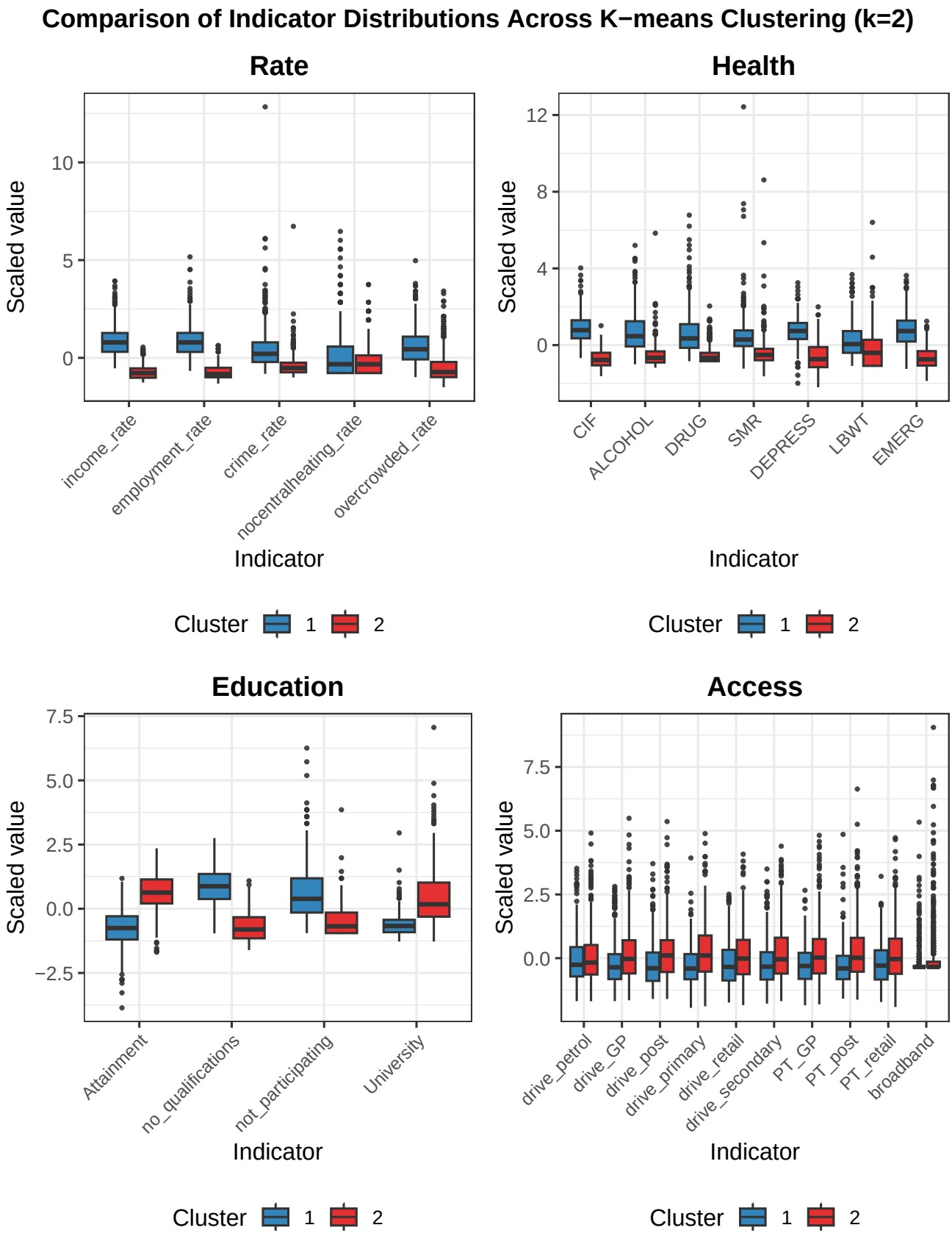
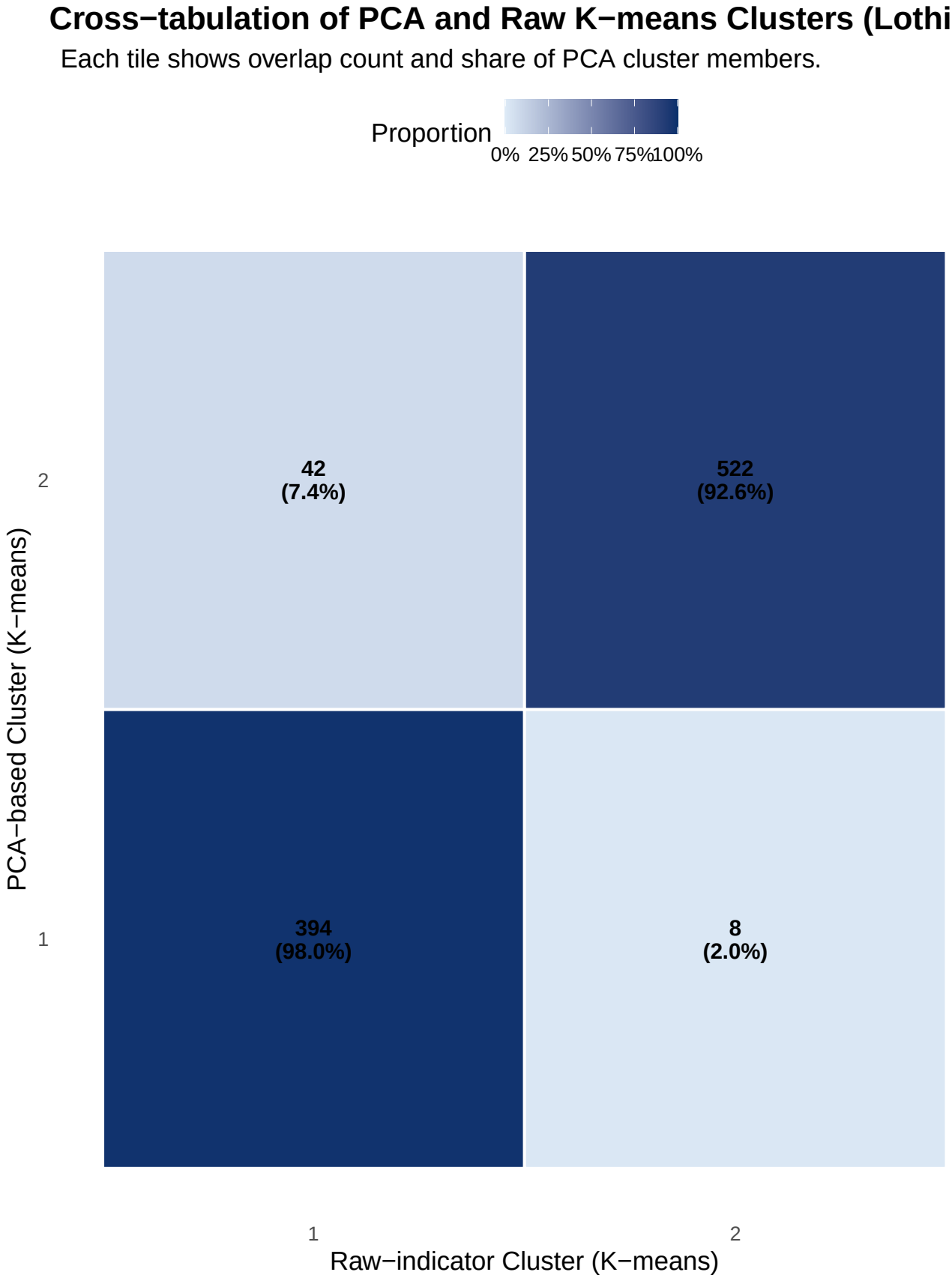
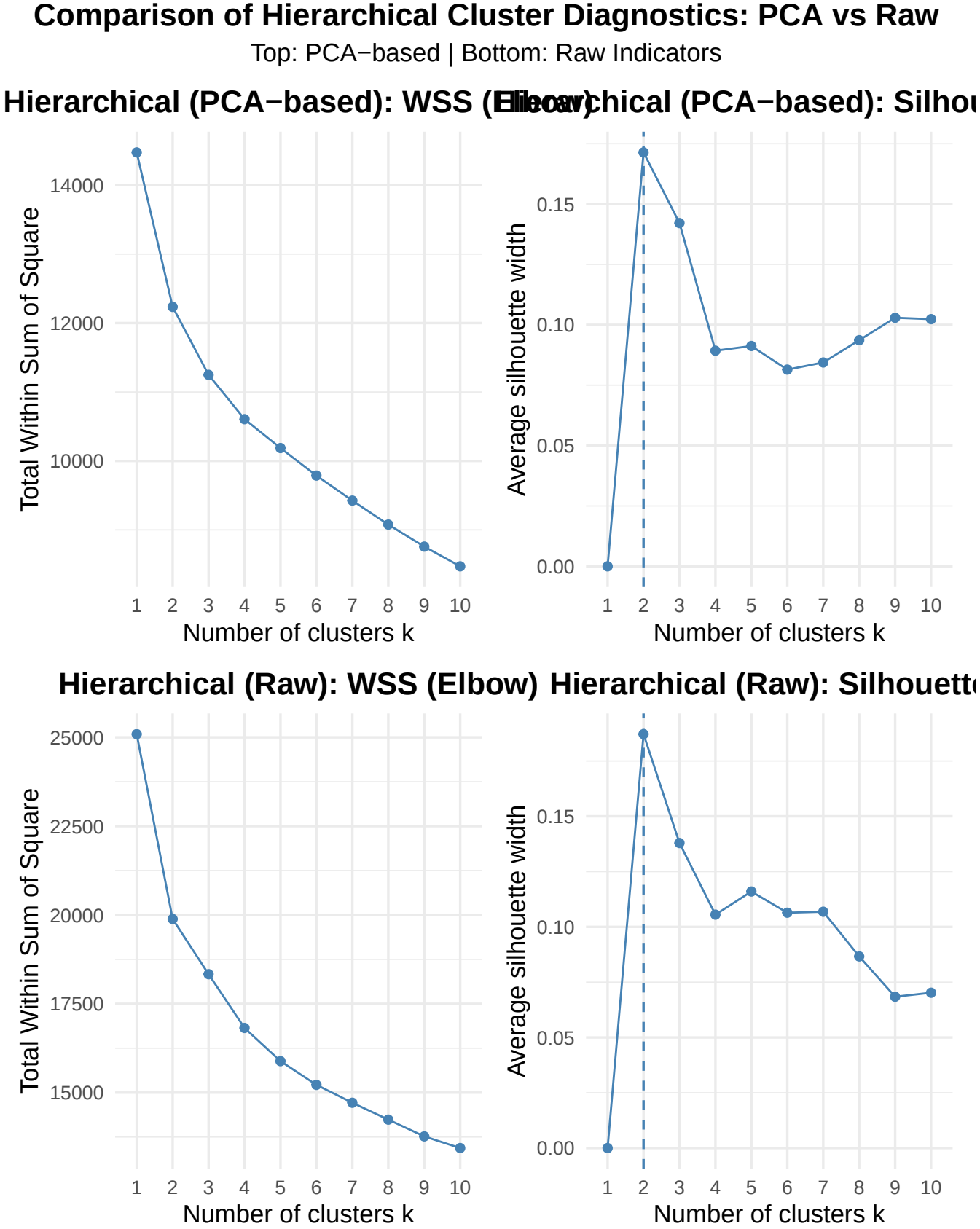

Comparison of Indicator Distributions Across K−means Clustering (k=2)
Rate
Scaled value
10
5
0
income_rate
employment_rate
crime_rate
nocentralheating_rate
overcrowded_rate
Indicator
Cluster 1 2
Health
Scaled value
12
8
4
0
CIF
ALCOHOL
DRUG
SMR
DEPRESS
LBWT
EMERG
Indicator
Cluster 1 2
Education
Scaled value
7.5
5.0
2.5
0.0
−2.5
Attainment
no_qualifications
not_participating
University
Indicator
Cluster 1 2
Access
Scaled value
7.5
5.0
2.5
0.0
drive_petrol
drive_GP
drive_post
drive_primary
drive_retail
drive_secondary
PT_GP
PT_post
PT_retail
broadband
Indicator
Cluster 1 2

Table 5: Similarity between PCA-based and Raw K-means clusters (k = 2)

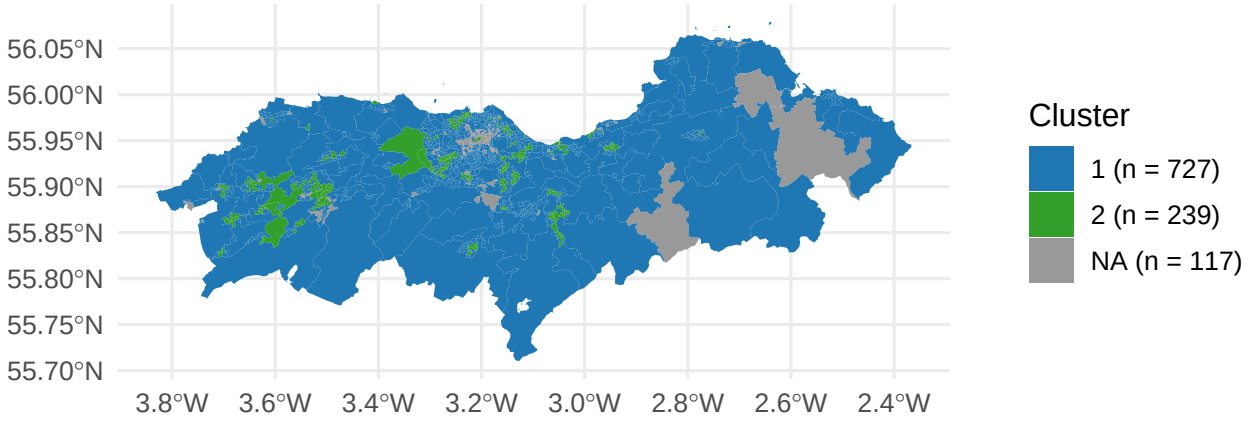
ARI	NMI	VI
0.803	0.716	0.55


Cross−tabulation of PCA and Raw K−means Clusters (Lothian)
Each tile shows overlap count and share of PCA cluster members.
Proportion
0%
25%
50%
75%
100%
42
(7.4%)
522
(92.6%)
394
(98.0%)
8
(2.0%)
PCA−based Cluster (K−means)
2
1
1
2
Raw−indicator Cluster (K−means)

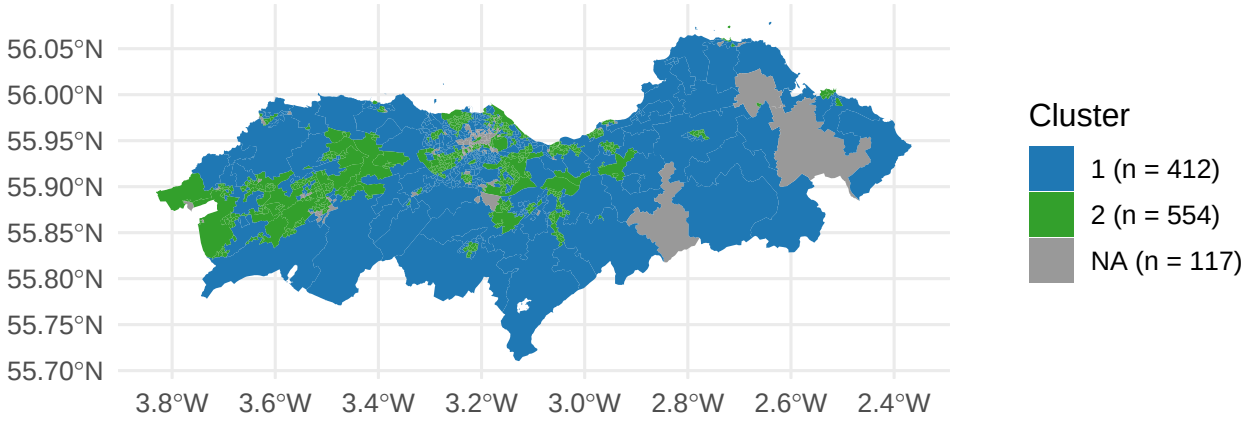

Comparison of Hierarchical Cluster Diagnostics: PCA vs Raw
Top: PCA−based | Bottom: Raw Indicators
Hierarchical (PCA−based): WSS (Elbow)
Hierarchical (PCA−based): Silhou
Total Within Sum of Square
Average silhouette width
Number of clusters k
Hierarchical (Raw): WSS (Elbow)
Hierarchical (Raw): Silhouett

Comparison of Hierarchical Clusters: PCA vs Raw

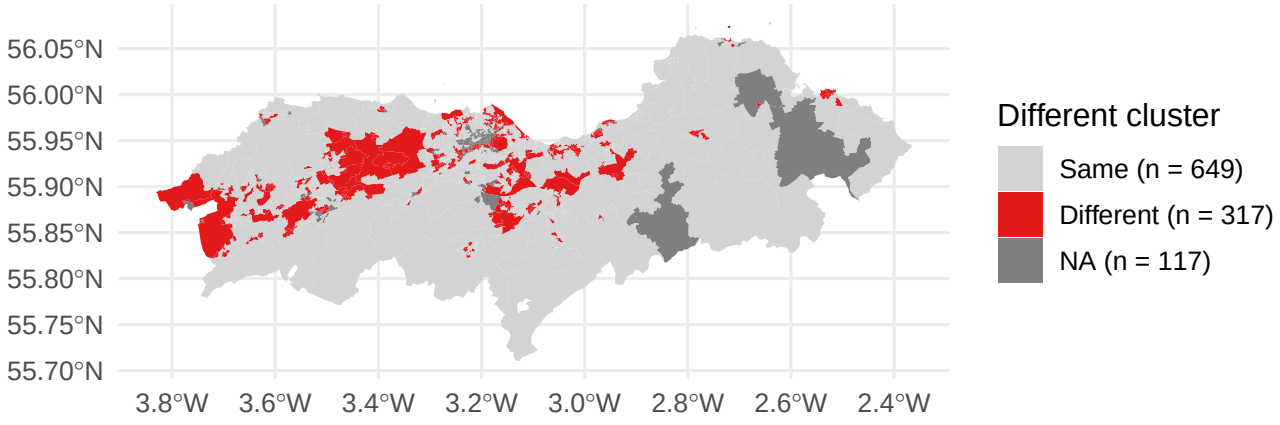
Hierarchical Clusters (PCA−based)

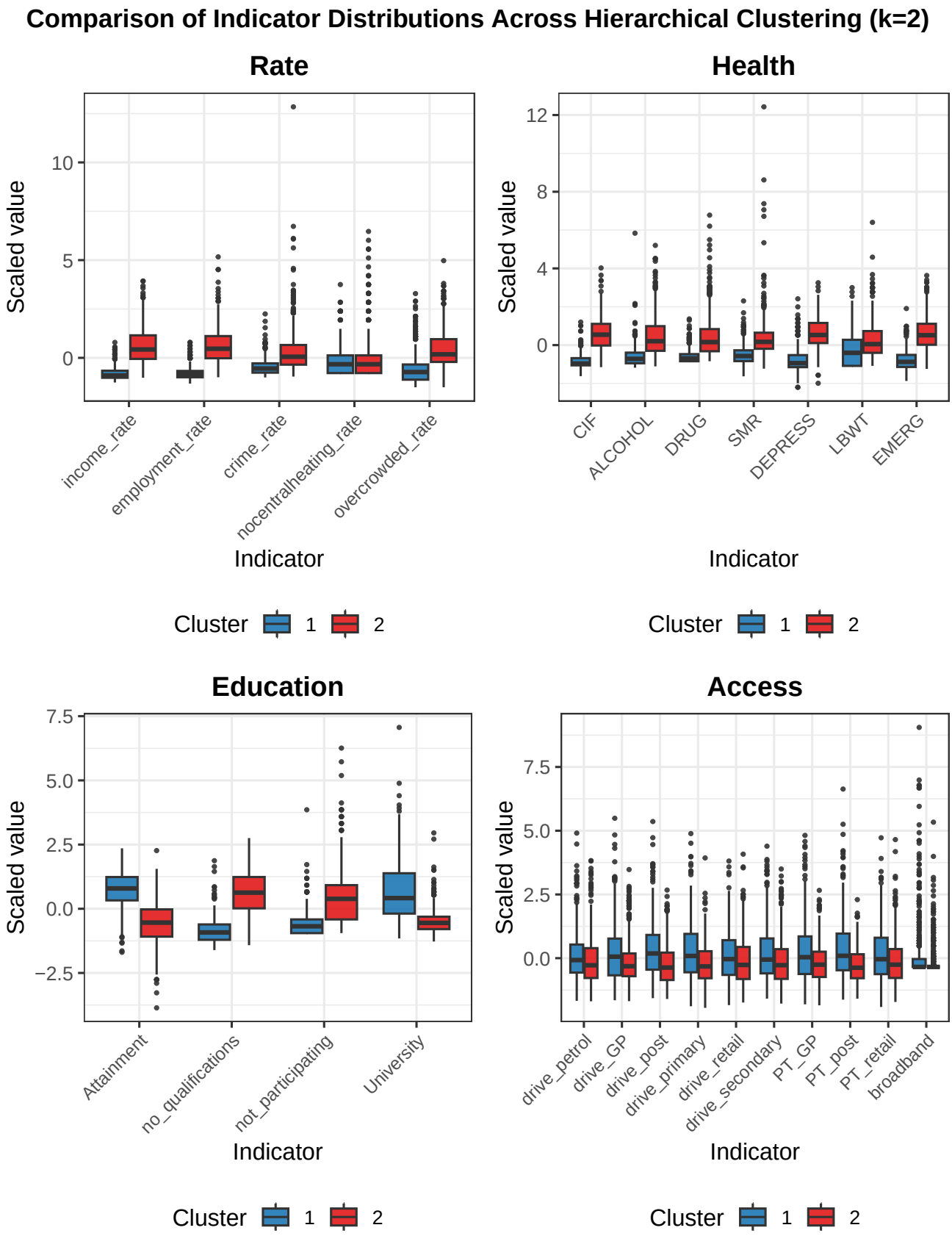


Hierarchical Clusters (Raw indicators)



Areas with Different Hierarchical Assignments (PCA vs Raw)




Comparison of Indicator Distributions Across Hierarchical Clustering (k=2)
Rate
Scaled value
10
5
0
income_rate
employment_rate
crime_rate
nocentralheating_rate
overcrowded_rate
Indicator
Cluster 1 2
Health
Scaled value
12
8
4
0
CIF
ALCOHOL
DRUG
SMR
DEPRESS
LBWT
EMERG
Indicator
Cluster 1 2
Education
Scaled value
7.5
5.0
2.5
0.0
−2.5
Attainment
no_qualifications
not_participating
University
Indicator
Cluster 1 2
Access
Scaled value
7.5
5.0
2.5
0.0
drive_petrol
drive_GP
drive_post
drive_primary
drive_retail
drive_secondary
PT_GP
PT_post
PT_retail
broadband
Indicator
Cluster 1 2

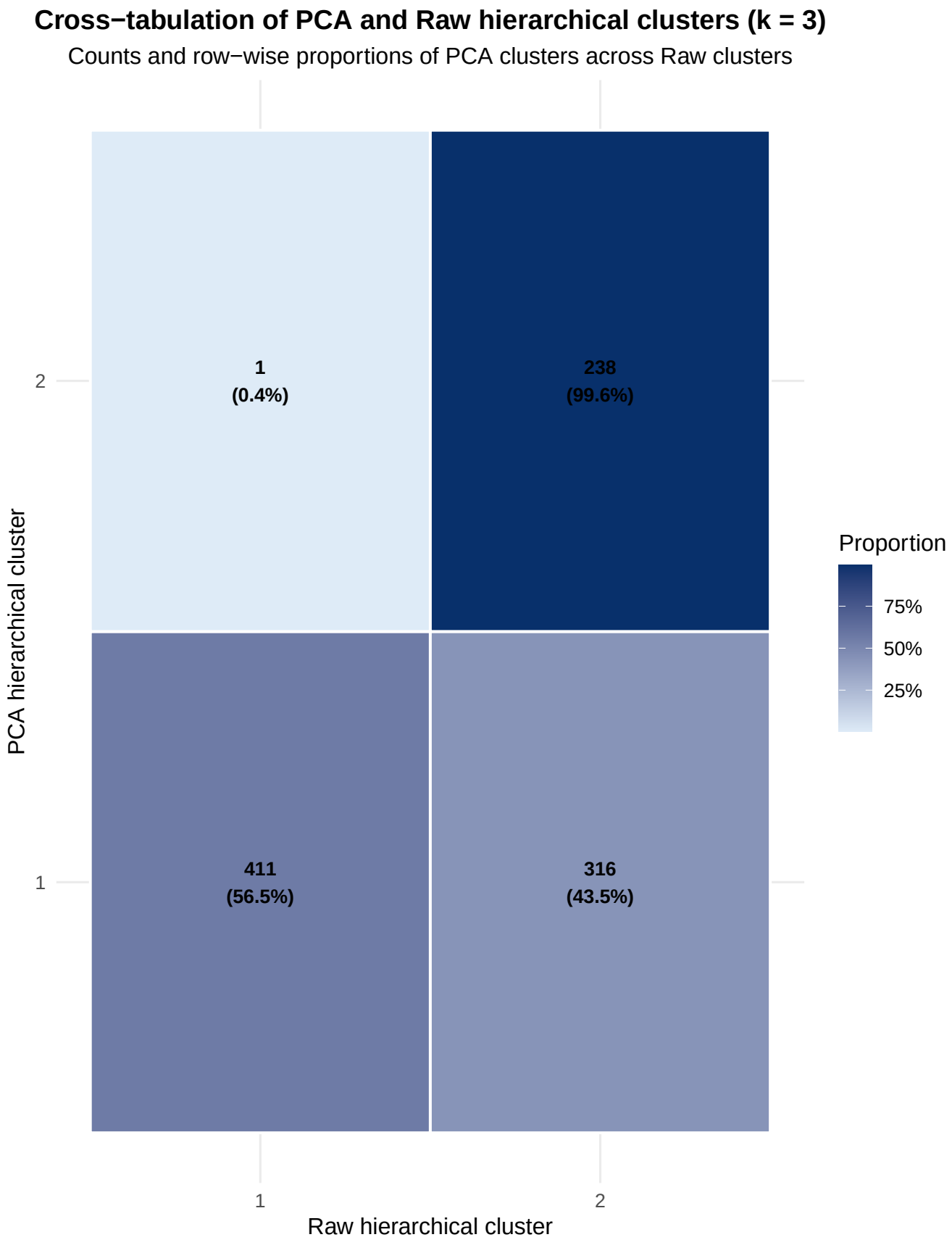
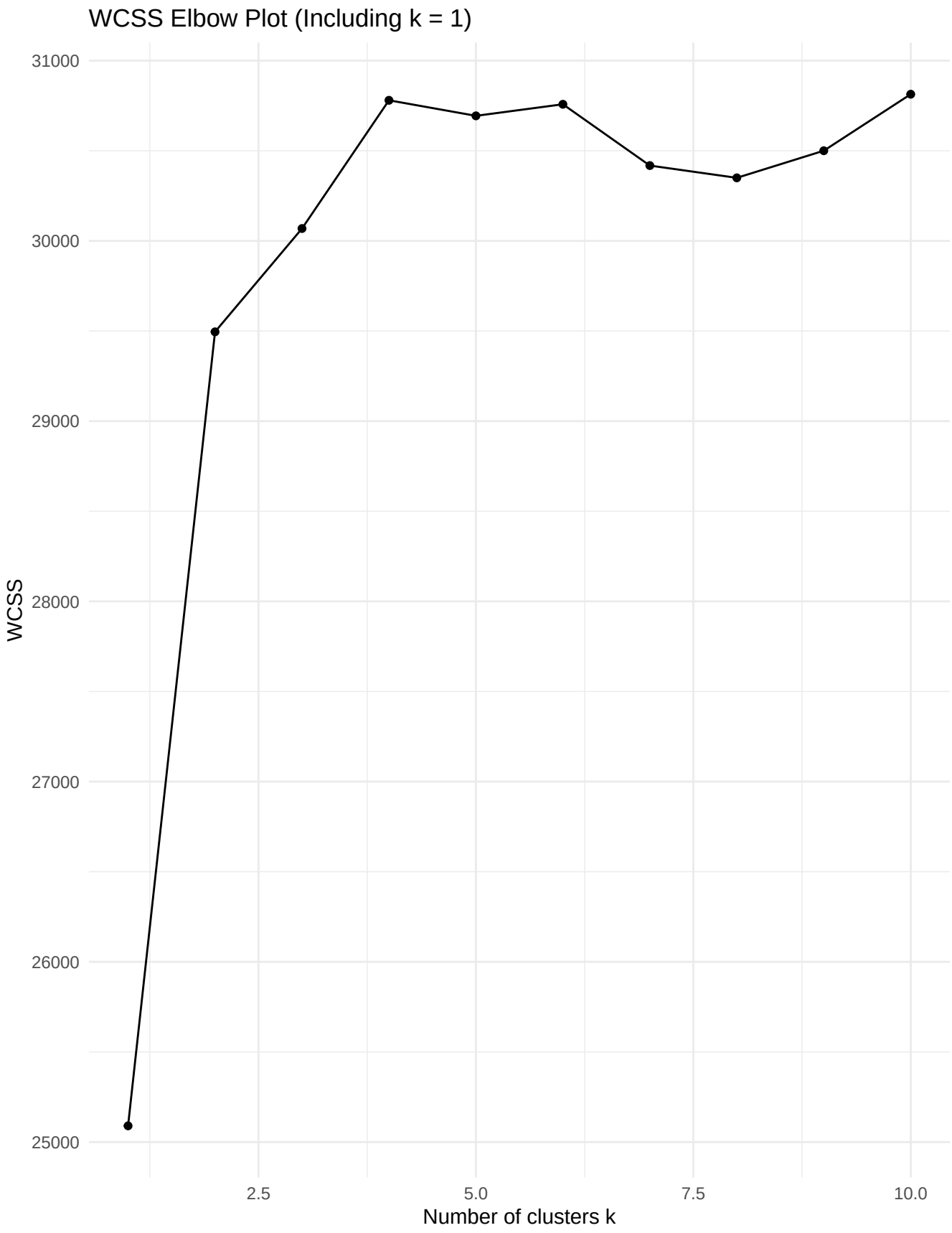
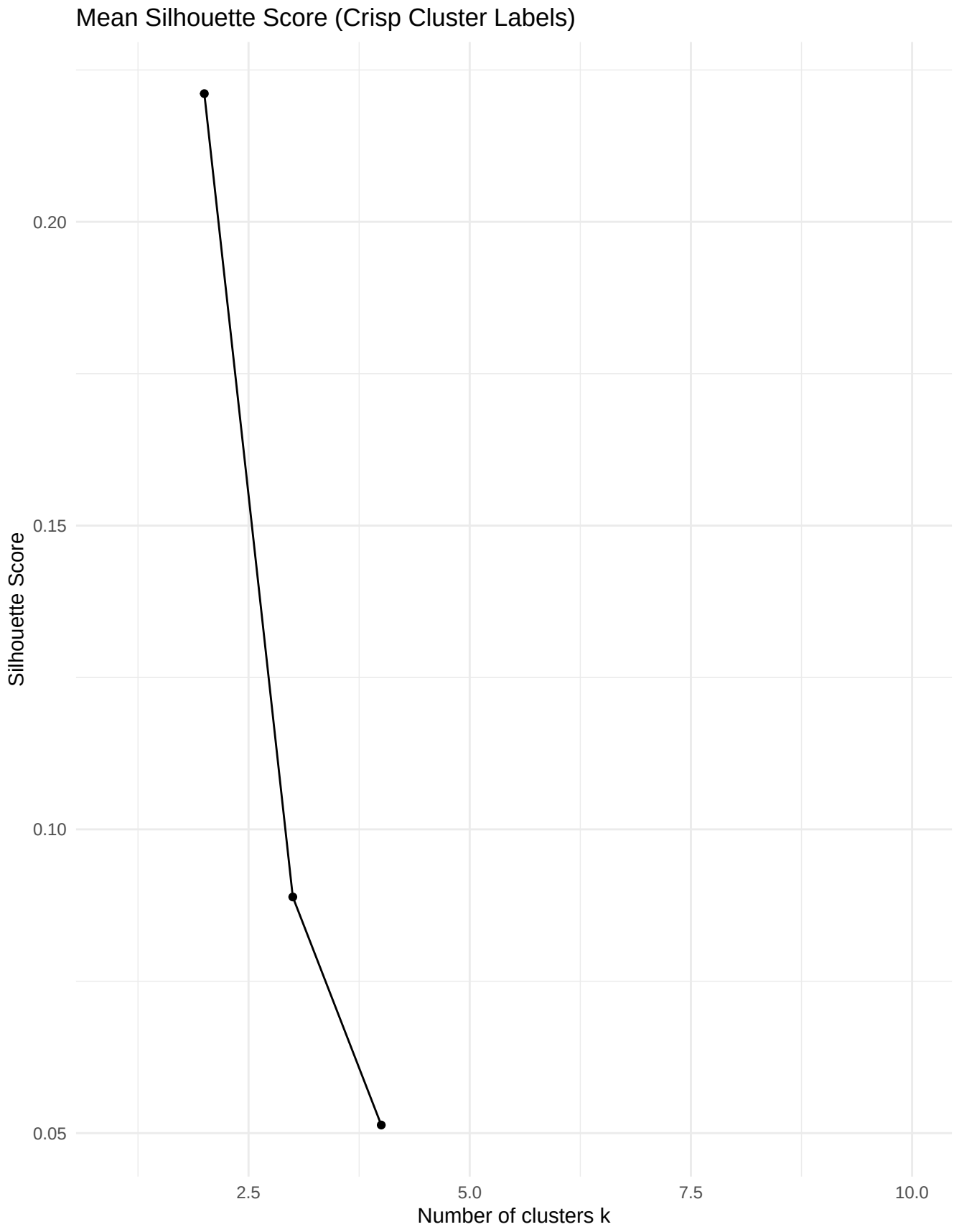

Cross−tabulation of PCA and Raw hierarchical clusters (k = 3)
Counts and row−wise proportions of PCA clusters across Raw clusters
1
(0.4%)
238
(99.6%)
411
(56.5%)
316
(43.5%)
PCA hierarchical cluster
2
1
Raw hierarchical cluster
1
2
Proportion
75%
50%
25%

Table 6: Similarity between PCA-based and Raw K-means clusters (k = 2)

ARI	NMI	VI
0.113	0.235	1.329


WCSS Elbow Plot (Including k = 1)
31000
30000
29000
28000
27000
26000
25000
WCSS
2.5
5.0
7.5
10.0
Number of clusters k


Mean Silhouette Score (Crisp Cluster Labels)
Silhouette Score
0.20
0.15
0.10
0.05
2.5
5.0
7.5
10.0
Number of clusters k

```
## # A tibble: 10 x 3
##        k   wcss silhouette
##    <int>  <dbl>      <dbl>
##  1     1 25090     NA
##  2     2 29495.     0.221
##  3     3 30069.     0.0889
##  4     4 30780.     0.0513
##  5     5 30694.    NA
##  6     6 30758.    NA
##  7     7 30418.    NA
##  8     8 30350.    NA
##  9     9 30500.    NA
## 10    10 30814.    NA
```

```
## # A tibble: 8 x 4
##       k   ARI   NMI    VI
##   <int> <dbl> <dbl> <dbl>
## 1     2 0.687 0.560 0.817
## 2     3 0.701 0.630 0.998
## 3     4 0.796 0.772 0.853
## 4     5 0.829 0.761 0.797
## 5     6 0.834 0.765 0.795
## 6     7 0.887 0.849 0.560
## 7     8 0.854 0.811 0.691
## 8     9 0.909 0.867 0.502
```

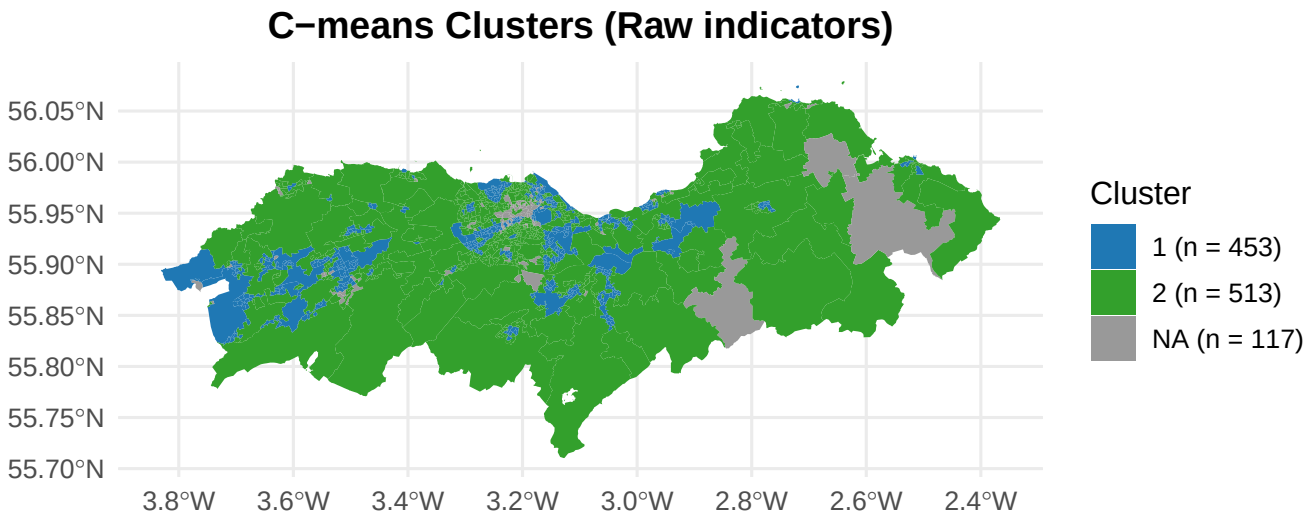
```
## # A tibble: 10 x 5
##        k empty_clusters avg_cluster_size min_cluster_size max_cluster_size
##    <int>          <int>            <dbl>            <int>            <int>
##  1     1              0             966              966              966
##  2     2              0             483              453              513
##  3     3              0             322              166              407
##  4     4              0             242.             127              368
##  5     5              0             242.             162              389
##  6     6              0             193.              65              379
##  7     7              0             193.               1              383
##  8     8              0             193.              12              393
##  9     9              0             193.              15              379
## 10    10              0             161                5              388
```

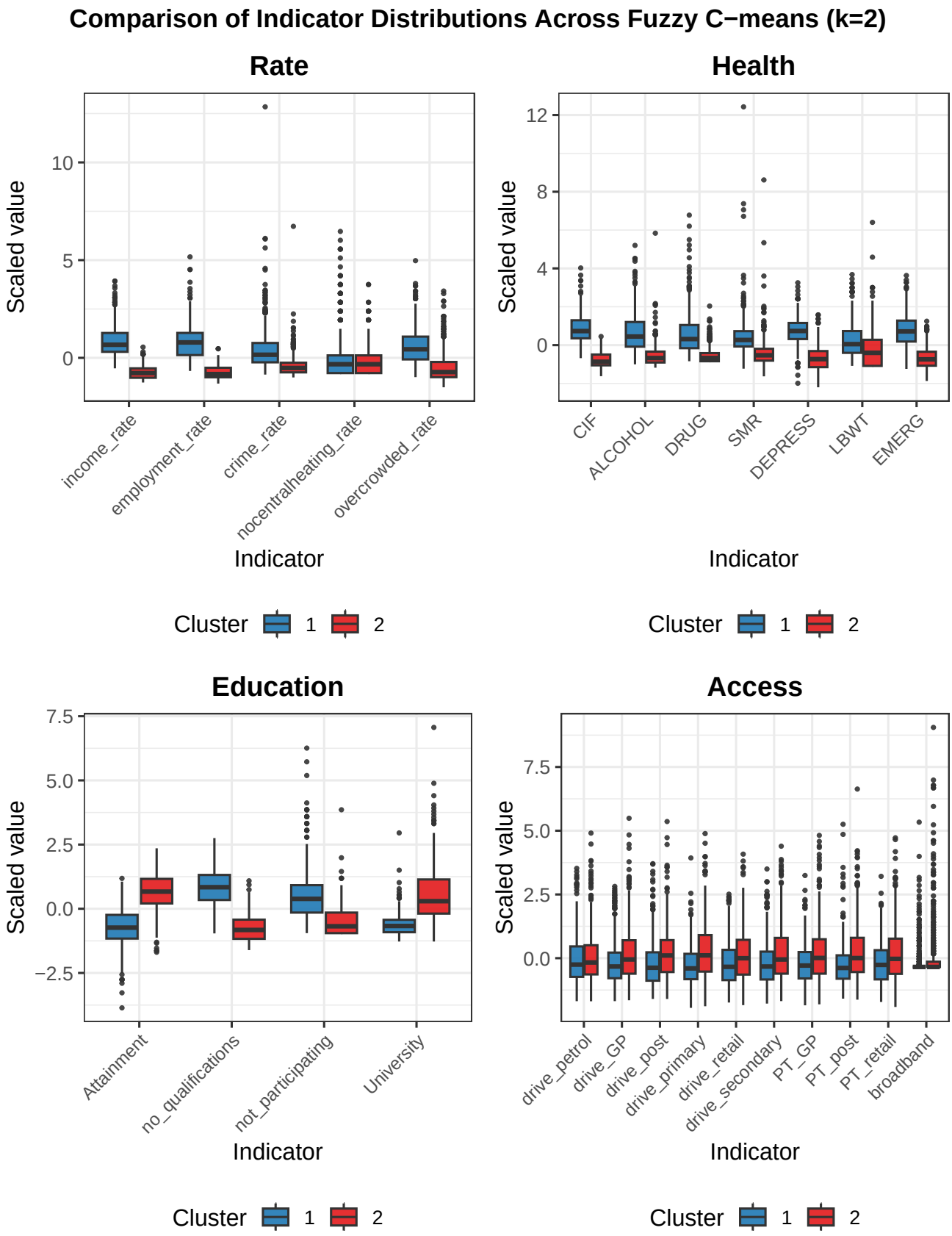
```
## # A tibble: 8 x 4
##       k   ARI   NMI    VI
##   <int> <dbl> <dbl> <dbl>
## 1     2 0.687 0.560 0.817
## 2     3 0.701 0.630 0.998
## 3     4 0.795 0.771 0.858
## 4     5 0.831 0.761 0.804
## 5     6 0.831 0.764 0.802
## 6     7 0.882 0.830 0.615
## 7     8 0.843 0.799 0.782
## 8     9 0.893 0.850 0.588
```

```
## # A tibble: 9 x 5
##       k      XB    PC    PE     FS
##   <int>   <dbl> <dbl> <dbl>  <dbl>
## 1     2 1.76e 0 0.562 0.629 18175.
```

```
## 2     3 6.40e 3 0.372 1.04  11082.
## 3     4 5.57e 9 0.281 1.32   7271.
## 4     5 1.02e 9 0.224 1.55   5930.
## 5     6 1.92e25 0.187 1.73   4668.
## 6     7 2.56e17 0.160 1.88   4515.
## 7     8 2.43e14 0.140 2.02   3877.
## 8     9 2.90e21 0.125 2.13   3611.
## 9    10 1.57e15 0.112 2.24   3435.
```

Xie–Beni (XB) – "compactness / separation" collapsed due to the small separation.




Comparison of Indicator Distributions Across Fuzzy C−means (k=2)
Rate
Scaled value
10
5
0
income_rate
employment_rate
crime_rate
nocentralheating_rate
overcrowded_rate
Indicator
Cluster 1 2
Health
Scaled value
12
8
4
0
CIF
ALCOHOL
DRUG
SMR
DEPRESS
LBWT
EMERG
Indicator
Cluster 1 2
Education
Scaled value
7.5
5.0
2.5
0.0
−2.5
Attainment
no_qualifications
not_participating
University
Indicator
Cluster 1 2
Access
Scaled value
7.5
5.0
2.5
0.0
drive_petrol
drive_GP
drive_post
drive_primary
drive_retail
drive_secondary
PT_GP
PT_post
PT_retail
broadband
Indicator
Cluster 1 2

Vine-Copula

Table 7: Similarity between Hierechical and K-means clusters (k = 2)

ARI	NMI	VI
0.528	0.502	0.981

Table 8: Similarity between c-means and K-means clusters (k = 2)

ARI	NMI	VI
0.923	0.872	0.252